Modeling Climate Stress Impacts on U.S. Corn Yield

Evaluating predictive robustness and regional differences

Georgetown University
M.S. in Data Science and Analytics
DSAN5550: Data Science for Climate Change

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Problem Statement

Can climate-based models trained in one region accurately predict crop yields in another?
Understanding generalization is critical for applying climate models under changing conditions. This project evaluates whether climate-yield relationships are stable across regions or region-specific.

Research Questions

RQ1: To what extent can growing-season climate stress variables explain variation in county-level corn yield within the U.S. Corn Belt?
RQ2: Do climate-yield models trained in the Corn Belt generalize to the Great Plains? And vice-versa?
RQ3: Do the dominant climate stressors associated with yield variability differ across these regions?

Data

- USDA NASS (yield), PRISM (temperature, precipitation), U.S. Drought Monitor (drought)
- Features: temperature, precipitation, drought
- ~21k county-year observations (2000-2022)
- Corn Belt (IL, IN, IA, MN, MO, OH, WI) vs Great Plains (KS, NE, ND, OK, SD, TX)

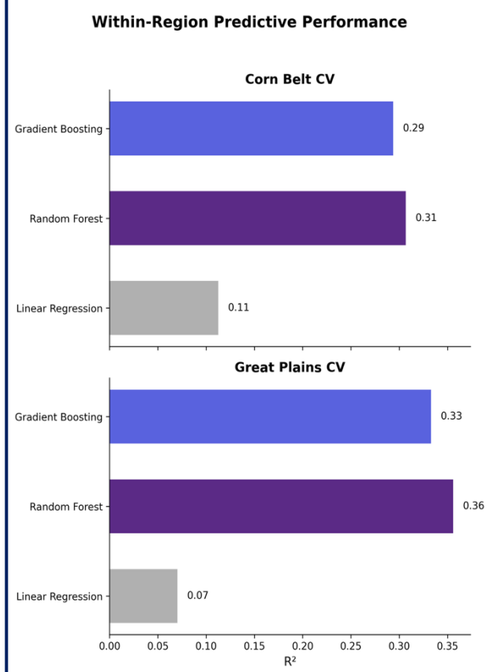
Methods

- Models: Linear Regression, Random Forest, Gradient Boosting
- Features: climate only vs climate + drought
- Metrics: RMSE, MAE, R^2
- Evaluation:
  - 5-fold cross-validation (within region)
  - Transfer: Corn Belt → Great Plains, and vice versa

Results

RQ1: Within-Region Prediction

- Models are evaluated separately within each region
- Performance is measured using cross-validation (R^2)

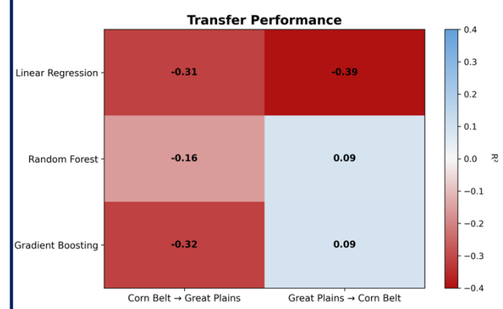


Nonlinear models outperform linear regression

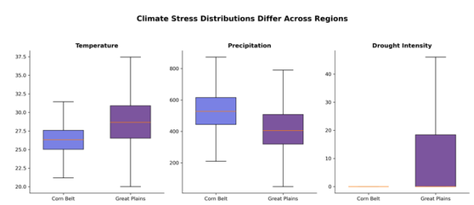
- Random Forest and Gradient Boost outperform linear regression
- Performance is moderate across both regions ($R^2 \approx 0.30$–0.36)
- Model performance is similar across regions

RQ2: Transfer Performance

- Models are trained in one region and evaluated in the other
- Transfer performance is measured using R^2



Models trained in one region FAIL to generalize to the other

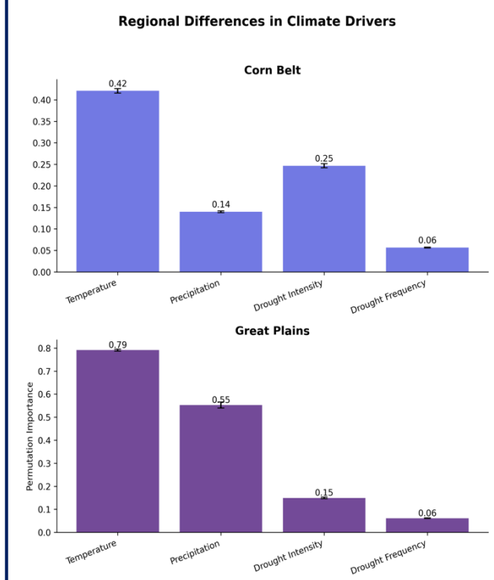


Climate conditions differ substantially between regions

- 4 of 6 transfer scenarios produce negative R^2 values
- Performance drops from 0.30 within-region to as low as -0.39 under transfer
- The Great Plains has higher temperatures, lower precipitation, and greater drought variability

RQ3: Feature Importance

- Feature importance is computed separately for each region
- Models are trained within-region (not transferred)



Models rely on different climate drivers across regions

- Temperature is the most important predictor in both regions
- Precipitation plays a much larger role in the Great Plains
- Drought intensity is more influential in the Corn Belt

Discussion

- Climate-yield relationships are region-specific and do not generalize across regions
- Transfer failure suggests models are sensitive to underlying data distributions
- Transfer asymmetry suggests models trained on more variable data generalize slightly better

Limitations

- Models do not account for temporal dynamics or lagged climate effects
- Only a limited set of climate variables is included, and potential confounding factors are not modeled
- Models assume relationships are stable within regions

Future Steps

- Develop region-aware or hierarchical models to improve cross-region generalization
- Incorporate additional environmental and management variables (soil, irrigation)
- Evaluate models under shifting climate conditions to assess robustness to climate change
- Extend the framework to identify climate-invariant predictors of yield

Carbon Cost

- Model training and evaluation runtime: **~1 minute (across all models)**
- Estimated CO_2 emissions: < 0.001 kg
- Does not include data processing or feature engineering steps

Data Sources:
USDA National Agricultural Statistics Service (NASS)
PRISM Climate Group, Oregon State University
U.S. Drought Monitor